影像處理
--CH3_2

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Q1.Please use a Histogram Statistics method and a local enhancement method to extract the hidden image of the image, 'hidden object.jpg.' Please describe the your parameters and method in detailand print out the source code?

1）Histogram Statistics method　source code：

```
import cv2
import numpy as np
import math
from pathlib import Path
from skimage.measure import shannon_entropy

INPUT_NAME = "hidden_object.jpg"
OUTPUT_NAME = "hsm_best.jpg"

def imread_unicode(path, flags=cv2.IMREAD_GRAYSCALE):
    p = Path(path)
    data = np.fromfile(str(p), dtype=np.uint8)
    return cv2.imdecode(data, flags)

def imwrite_unicode(path, img, ext=".jpg"):
    ok, buf = cv2.imencode(ext, img)
    Path(path).parent.mkdir(parents=True, exist_ok=True)
    buf.tofile(str(path))

def var(pdf, mu):
    return sum(((i - mu) ** 2) * pdf[i] for i in range(len(pdf)))

def mean(pdf):
    return sum(i * pdf[i] for i in range(len(pdf)))

def hist(img, dim, k0, k1, k2, k3, e):
    assert dim % 2 != 0
    height, width = img.shape[:2]
    img_pdf = np.zeros(256, dtype=np.float64)
    img_flat = img.ravel()
    for v in img_flat:
        img_pdf[int(v)] += 1.0
    img_pdf /= img_flat.size
```

```
    img_mean = mean(img_pdf)
    print(f"Global mean = {img_mean:.3f}")
    img_sd = math.sqrt(var(img_pdf, img_mean))
    print(f"Global std  = {img_sd:.3f}")
    local_area = dim * dim
    r = dim // 2
    out = img.copy()
    local_mean_img = np.full(img.shape, np.nan, dtype=np.float64)
    local_std_img  = np.full(img.shape, np.nan, dtype=np.float64)
    step = 50
    max_print = 20
    printed = 0
    for i in range(r, height - r):
        for j in range(r, width - r):
            local_pdf = np.zeros(256, dtype=np.float64)
            for q in range(i - r, i + r + 1):
                for p in range(j - r, j + r + 1):
                    local_pdf[int(img[q, p])] += 1.0 / local_area
            lm = mean(local_pdf)
            ls = math.sqrt(var(local_pdf, lm))
            local_mean_img[i, j] = lm
            local_std_img[i, j]  = ls
            if printed < max_print and ((i - r) % step == 0) and ((j -
r) % step == 0):
                printed += 1
    core_mean = local_mean_img[r:height - r, r:width - r]
    core_std  = local_std_img [r:height - r, r:width - r]
    core_img  = img           [r:height - r, r:width - r]
    mask_core = (
        (core_mean >= img_mean * k0) & (core_mean <= img_mean * k1) &
        (core_std  >= img_sd   * k2) & (core_std  <= img_sd   * k3)
    )
    print("Overall range of the original image:", int(img.min()), "to",
int(img.max()))
    if np.any(mask_core):
        sel = core_img[mask_core]
        print("Enhanced area range:", int(sel.min()), "to",
int(sel.max()))
```

```
    out_core = out[r:height - r, r:width - r]
    out_core[mask_core] = np.clip(out_core[mask_core] * e, 0, 255)
    return out.astype(np.uint8)

if __name__ == "__main__":
    here = Path(__file__).resolve().parent
    in_path = here / INPUT_NAME
    img1 = imread_unicode(in_path, cv2.IMREAD_GRAYSCALE)
    k0 = 0.0
    k1 = 0.10
    k2 = 0.0
    k3 = 0.06
    e  = 22.0
    out = hist(img1, dim=3, k0=k0, k1=k1, k2=k2, k3=k3, e=e)
    out_path = here / OUTPUT_NAME
    imwrite_unicode(out_path, out, ext=".jpg")
    print(f"\nOutput：{out_path}")
```

2)depict　Histogram Statistics method:

用全域與局部的平均與標準差做門檻判斷，只增強夠暗且對比夠低的區域，以把隱藏在暗部的圖樣拉出來。判斷式為$K_0 m_G \le m_{Sxy} \le K_1 m_G$與$K_2 \sigma_G \le \sigma_{Sxy} \le K_3 \sigma_G$，符合者將像素乘上一個係數e。

3)depict Histogram Statistics method parameter:

本次 LAB 的目的在於從hidden_object的五個黑色方格中提取隱藏圖案，因此這些方格可視為非常暗的區域。依據公式與程式計算結果，全域平均灰階值約為m_G=161。若設定K_1=0.1與K_0=0，則條件為$m_{Sxy} \le K_1 m_G = 161 * 0.1 = 16.1$，也就是僅選取局部均值低於 16.1 的區域，確保提取範圍集中在極暗區塊。

全域標準差計算結果約為σ_G=103。若設定K_3=0.06與K_2=0，則條件為$\sigma_{Sxy} \le K_3 \sigma_G = 0.06 * 103 = 6.2$，這表示僅增強對比度極低的平坦暗區域，可避免邊界或具有紋理的高對比區塊被誤判並放大。

最後，原始圖像中被選取的像素灰階值範圍大約落在 0 至 11 之間。為了將其有效提升至接近全域的亮度範圍（0 ~ 228），選擇增益倍率e=$\frac{228}{11}$=22.0，如此能將暗部訊號強化到接近可見範圍，讓隱藏的圖案更為清晰。

4) Histogram Statistics method result:

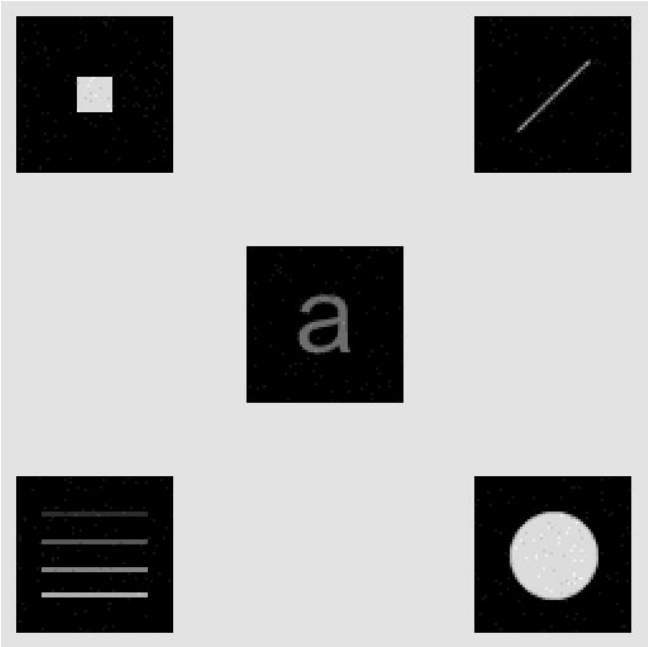

Figure1. Histogram Statistics method result

```
Global mean = 160.656
Global std  = 102.998
Overall range of the original image: 0 to 228
Enhanced area range: 0 to 11
```

Figure2. parameter design reference

5) local enhancement method source code:

```
mport cv2
import numpy as np

def apply_gamma(img, gamma):
    table = np.array([(i / 255.0) ** gamma * 255 for i in
range(256)]).astype("uint8")
    return cv2.LUT(img, table)

def method_c(img, gamma=0.5, clip=8.0, tiles=4):
    img_g = apply_gamma(img, gamma)
    clahe = cv2.createCLAHE(clipLimit=clip, tileGridSize=(tiles, tiles))
    return clahe.apply(img_g)

def post_ab(img, alpha=1.4, beta=30):
    return cv2.convertScaleAbs(img, alpha=alpha, beta=beta)
```

```python
def main():
    input_file = "hidden_object.jpg"
    output_file = "LE_result.png"
    img = cv2.imread(input_file, cv2.IMREAD_GRAYSCALE)
    base = method_c(img, gamma=0.2, clip=8.0, tiles=1)
    result = post_ab(base, alpha=1.4, beta=30)
    cv2.imwrite(output_file, result)
    cv2.imshow("Enhanced", result)
    cv2.waitKey(0)
    cv2.destroyAllWindows()
if __name__ == "__main__":
    main()
```

6) depict local enhancement method:

採用 Gamma 校正先將暗部提亮，再利用 CLAHE 進行局部直方圖均衡化以強化低對比細節，最後透過 Alpha/Beta 微調整體亮度與對比度，使黑色方格內的圖案更加清晰可見。

7) depict local enhancement method parameter:

Gamma公式為I_{out}=255 × $(\frac{I_{in}}{255})^{\gamma}$，若$\gamma$<1，則暗部變亮，反之，則暗部變暗。採用gamma=0.2避免影像過度亮化，同時有效提升黑色區域的細節可見度。

在CLANE中，clipLimit為對比限制因子，數值越大則對比提升效果越強，暗部細節會更明顯，但伴隨著雜訊也可能放大。為兼顧亮度提升與避免過度曝光，採用 clipLimit = 8.0。tileGridSize為分塊大小，數值越小則越局部，暗塊會被更積極地處理，因此採用(1,1)。

最後α和β遵循的公式為I_{out}=αxI_{in}+β，其中α控制對比度，β控制亮度。在設定上α通常介於1.2～1.6間，因此採用中間值1.4。β通常介於20～45間，因此採用中間值30，以達到適度增強而不致失真。

8) local enhancement method result:

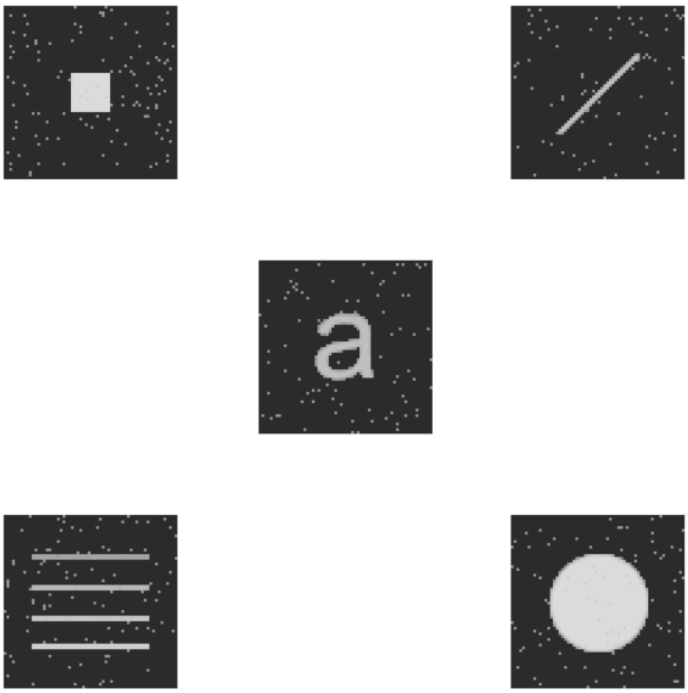

Figure3. local enhancement method result

Q2.Please comment and compare Histogram Statistics method and a local enhancement method?

Histogram Statistics適合整體影像灰階分布均勻的情況，能在不產生過多雜訊的前提下改善對比。而Local Enhancement 更適合用在局部區域對比不足的影像，但要注意控制參數（clipLimit、tileGridSize），否則會犧牲平滑性而導致噪聲。

Histogram Statistics 對比度提升有限，暗區細節顯得模糊、不夠清楚。Local Enhancement 可以把暗區細節拉得更清楚，但缺點是容易引入雜訊顆粒（Figure3 背景中出現白點斑駁現象）。